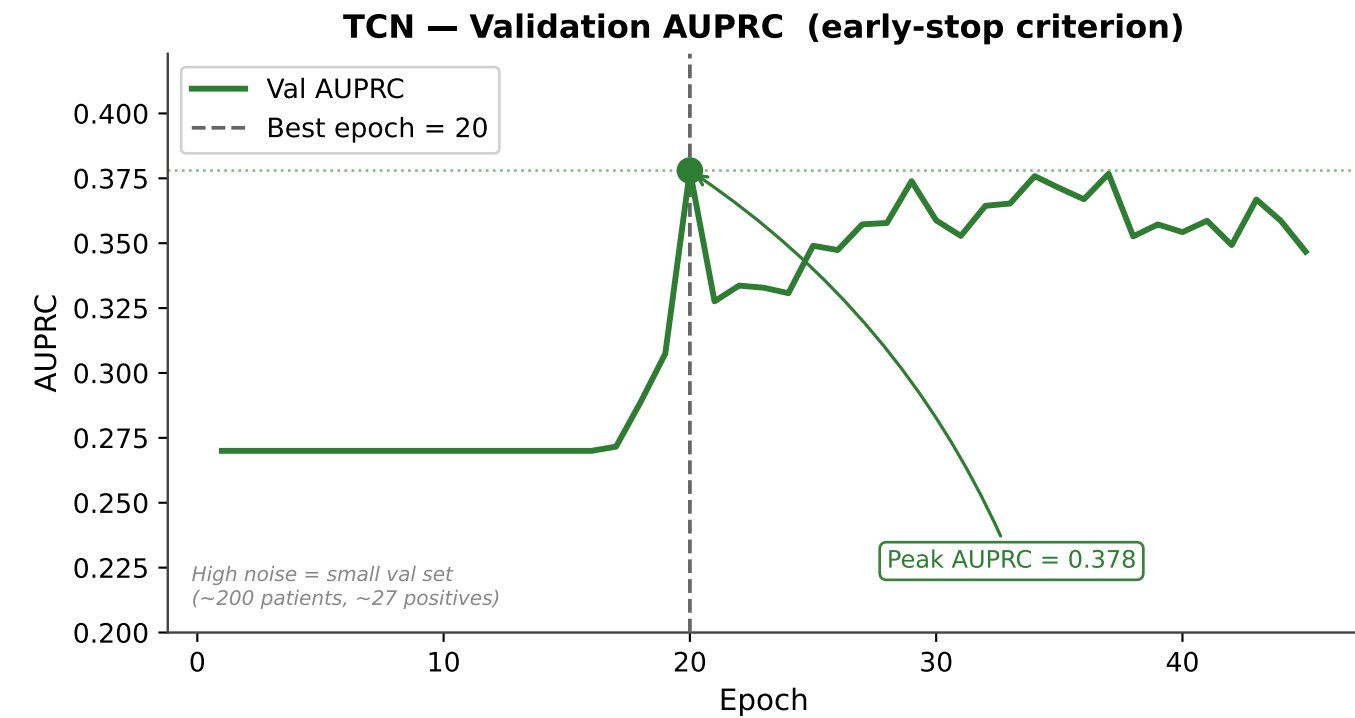
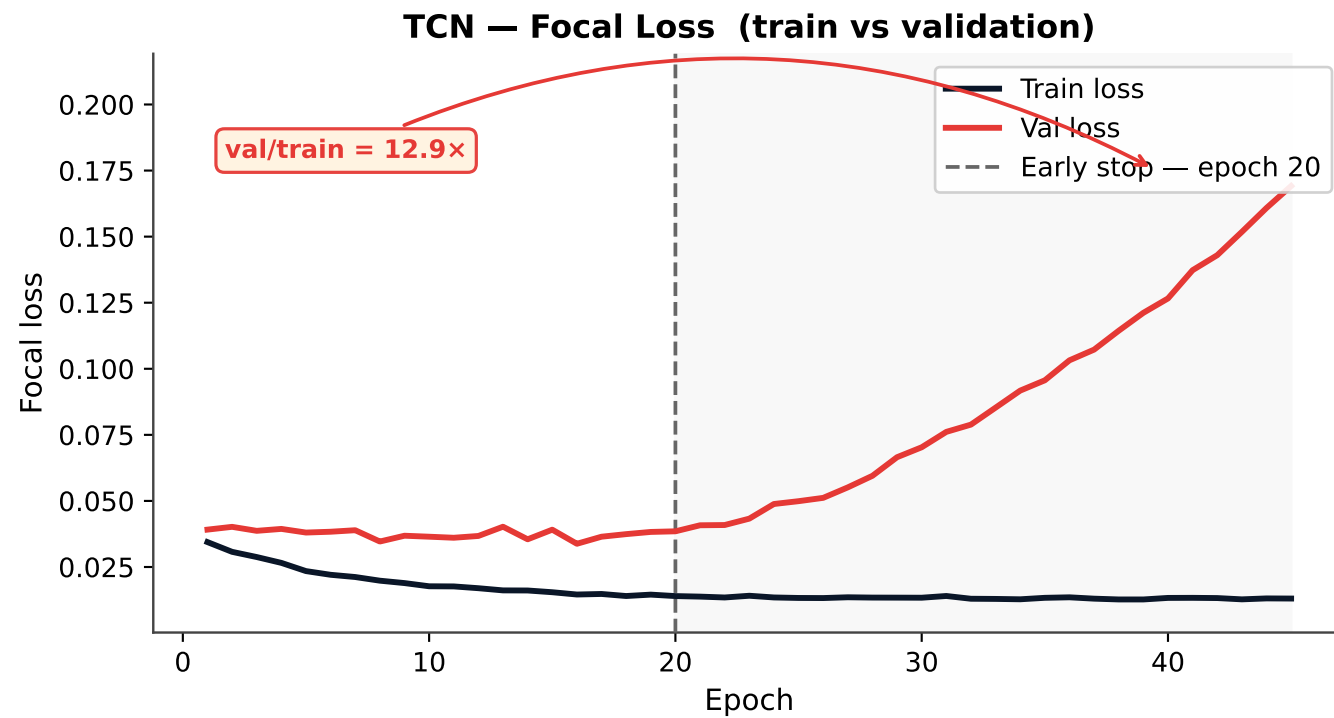
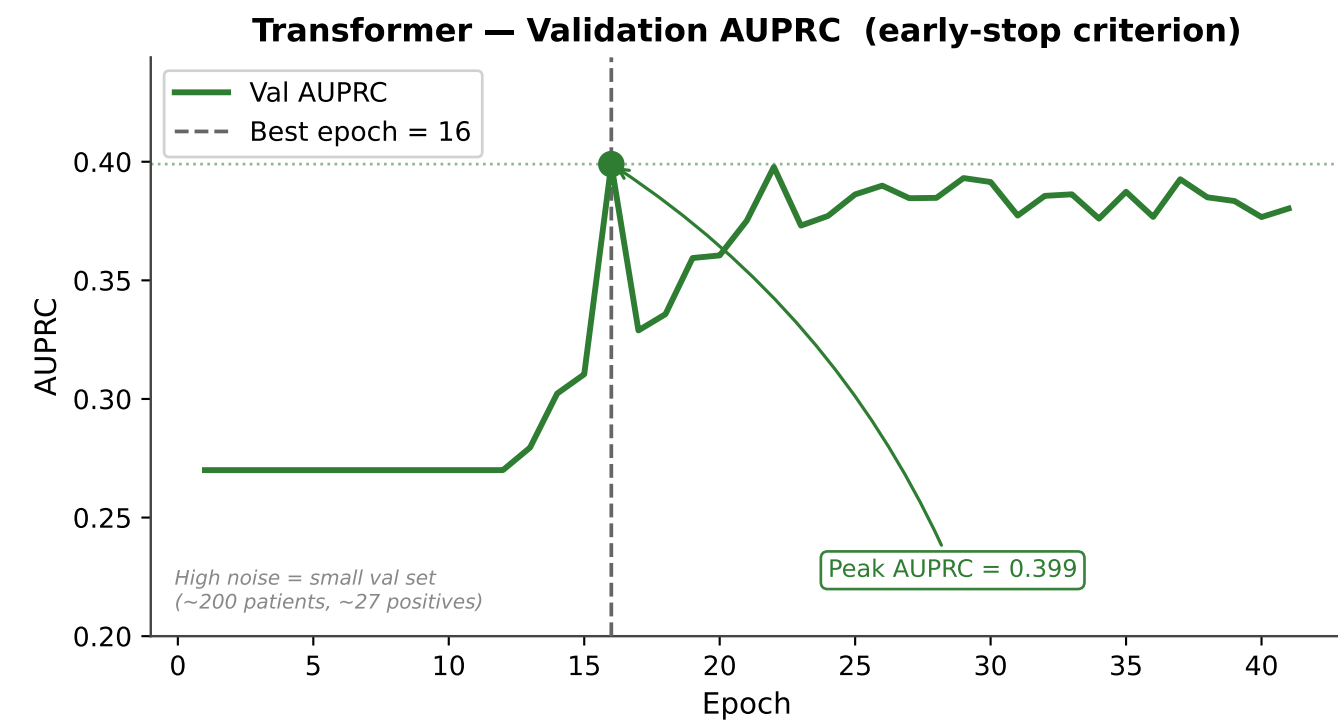
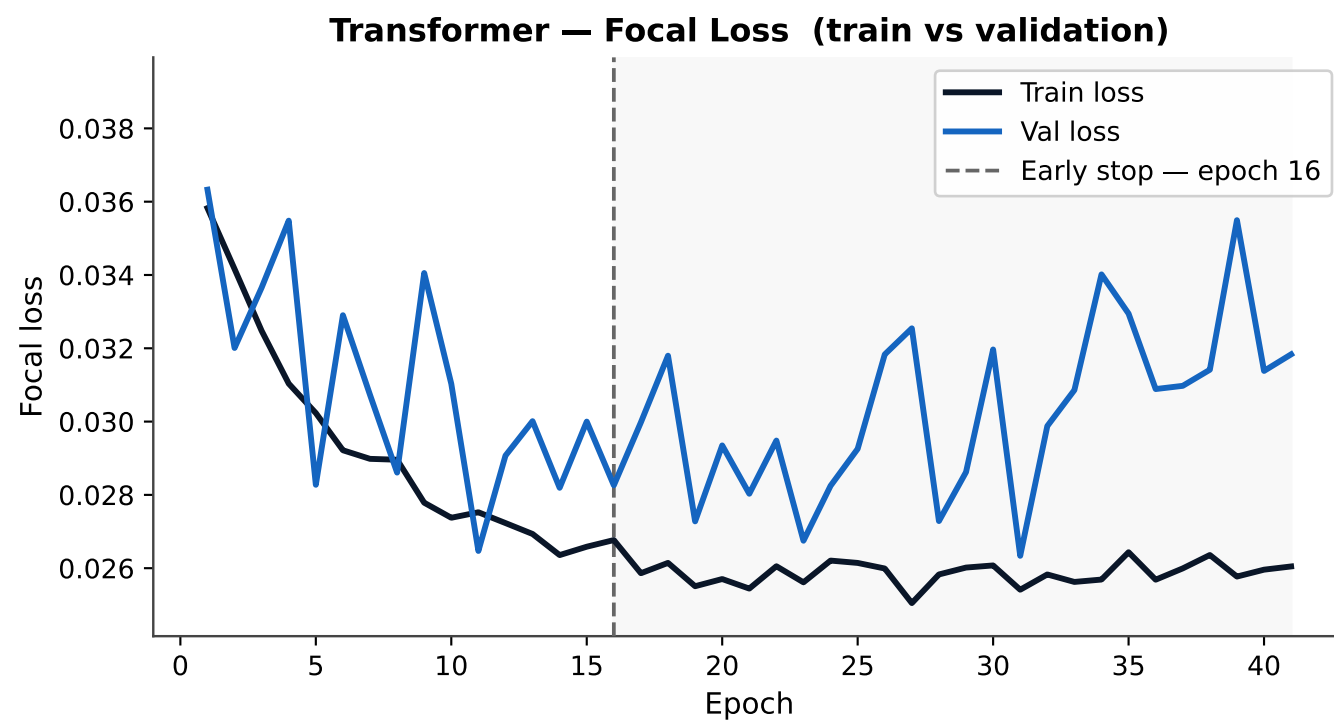


Figure 4.1 — Training & Validation Curves: TCN and Time-Aware Transformer
Focal loss (a=0.86, y=2.0) | Early stopping on val AUPRC | patience=25 | lr=3e-4 | batch=64



Diagnosis: Val loss diverges ($\times 7.7$ vs train at epoch 45) | Calibration collapsed: HL-H=205.8 | AUROC=0.819 (discrimination OK) | Root cause: focal loss ($\gamma=2$) amplifies gradient on hard examples + small train set (~2800 patients)



Diagnosis: Val loss tracks train loss ($\times 1.5$ at epoch 41) — stable training | CLS bottleneck + pre-LayerNorm act as implicit regularisation | AUROC=0.854 (best overall) | HL-H=165.2 (moderate miscalibration from focal loss)